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Guided Research

Calibration of Multi-output Gaussian Processes for Astronomical Light Curve Modeling

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Abstract

Gaussian processes (GPs) are widely used for reconstructing sparse and irregular astronomical light curves because they provide both predictive estimates and associated uncertainties. However, the calibration of their predictive distributions remains challenging, particularly when observations are sparse and multiple photometric bands are jointly modeled.

In this study, we investigate the predictive accuracy and uncertainty calibration of single-output and multi-output GPs using PLAsTiCC light curves. We evaluate their performance using RMSE, NLPD, CRPS, empirical coverage, standardized residuals, and PIT/KS-PIT diagnostics. We further examine how target-band data sparsity, light-curve characteristics, and cross-band similarity affect reconstruction performance.

The results show that wavelength-dependent multi-output GPs generally outperform single-band and independent-band baselines, particularly when target-band observations are sparse. Nevertheless, substantial negative transfer occurs for some objects, where differences in flux amplitude between bands can lead to unrealistic peak predictions. To address this issue, we investigate a band-specific amplitude kernel that allows different bands to have different marginal variances. This modification mitigates several severe failure cases, but does not improve the overall predictive performance compared with the original kernel.

1 Introduction

1.1 Motivation

An astronomical light curve shows how the brightness of a celestial object changes over time. It is inherently sparse and irregularly sampled, with observations often distributed unevenly across different photometric bands. These characteristics pose challenges for downstream analysis. Feature-based machine-learning methods like classification and regression trees typically rely on quantities extracted independently from individual bands [1, 2, 3], whose reliability can deteriorate when the observations in a band are sparse or missing. Although modern deep-learning methods can learn representations directly from irregular time series, their predictions are generally less directly interpretable and reliable uncertainty quantification remains challenging [4]. Consequently, reconstructing sparsely observed light curves remains an important preprocessing step for both direct scientific analysis and downstream machine-learning applications.

GP is a stochastic process that have become a widely used approach for light-curve reconstruction in time-domain astronomy. Unlike interpolation methods that provide only point estimates, GPs define a Gaussian probability distribution over functions and therefore provide both a predictive mean and an associated predictive uncertainty at unobserved times. Their ability to operate directly on irregularly sampled observations makes them particularly suitable for astronomical time series. For example, the winning solution of the PLAsTiCC classification challenge reconstructed light curves using GPs with a Matérn-3/2 kernel before extracting features for classification [5]. GPs have also been applied to supernova light curves and spectra: Kim et al. [6] used GPs to predict absolute magnitudes of the Type Ia supernovae (SNe Ia) by modeling the covariance between absolute magnitudes of different supernovae with a parameterized function that depends on differences between light-curve shapes and colors. Saunders et al. [7] modeled the full spectral time-series of SNe la with GPs. Léget et al. [8] used GPs to interpolate and project the observed spectra onto a common time grid.

The probabilistic nature of GPs is particularly attractive when observations are sparse, since the uncertainty of a reconstructed light curve can in principle reflect how strongly a prediction is supported by the available observations. However, the predictive distribution produced by a GP is not necessarily well calibrated. Its uncertainty depends on assumptions about the kernel, observational noise, hyperparameter estimation, and the temporal distribution of observations. A model may therefore achieve accurate point predictions while being systematically overconfident or underconfident. This distinction is important in astronomical applications where reconstructed light curves and their uncertainties may subsequently be used for physical inference or downstream classification.

Multi-band observations provide an additional opportunity to improve reconstruction. Astronomical transients are observed through multiple photometric filters, and their temporal evolution is often correlated across bands. A multi-output Gaussian process (MOGP) can exploit these correlations by jointly modeling observations from multiple bands, allowing information from a well-observed auxiliary band to constrain predictions in a sparsely observed target band. However, additional cross-band information does not necessarily improve the predictive distribution. If the assumed covariance between bands does not adequately represent their joint evolution, information transferred from auxiliary bands may introduce prediction errors or reduce predictive uncertainty without a corresponding improvement in accuracy, resulting in miscalibrated predictions.

1.2 Research Questions and Contributions

This work investigates the predictive accuracy and uncertainty calibration of Gaussian process (GP) models for multi-band astronomical light-curve reconstruction using the PLAsTiCC dataset. We compare single-band GPs with multi-output GP approaches and evaluate both point prediction accuracy and probabilistic performance using root mean squared error (RMSE), negative log predictive density (NLPD), continuous ranked probability score (CRPS), empirical prediction-interval coverage, standardized residuals, and probability integral transform (PIT/KS-PIT) diagnostics.

In particular, this study addresses three main research questions:

1. How do observation sparsity and light-curve characteristics affect the predictive accuracy and uncertainty calibration of GPs?

2. Under which conditions does wavelength-dependent cross-band information improve target-band reconstruction, and how do these benefits depend on target-band sparsity, auxiliary-band selection, and cross-band similarity?

3. What factors contribute to negative cross-band information transfer, and can a more flexible covariance kernel mitigate these failures without compromising overall predictive performance?

To address these questions, we first analyze the calibration of single-band GPs across different astronomical object classes and light-curve characteristics. We then conduct controlled comparisons between single-band, independent-band, and wavelength-dependent multi-output GPs to distinguish the effects of shared hyperparameter estimation and cross-band covariance. Finally, we investigate representative negative-transfer cases and introduce a band-specific amplitude kernel to examine the trade-off between covariance flexibility and effective cross-band information sharing.

Together, these experiments provide an empirical assessment of when multi-band GP reconstruction is beneficial, where its limitations arise, and how covariance kernel design influences predictive performance and uncertainty calibration.

2 Dataset

2.1 PLAsTiCC light curves

The PLAsTiCC dataset contains simulated transient data observed by the Large Synoptic Survey Telescope (LSST) under realistic observing conditions [9]. It consists of 7848 light curves of 14 different kinds of spectroscopically confirmed objects with photometric bands u, g, r, i, z. The observation time, flux, and flux uncertainty are provided. Its sampling is irregular, thus suitable for GP analysis. Flux data is available across all bands at each observed timestamp.

2.2 Experimental subset

We only take first 3000/300 light curves used due to computational cost. The number of objects from each class is listed in Table 1. The experiment subset with 3000/300 objects has similar class distribution with the full dataset.

2.3 Light-curve Characteristics

Astronomical light curves exhibit diverse temporal patterns across different object classes, while observations from different photometric bands often share similar temporal structures. To illustrate these characteristics, we first examine the r-band light curves from all 14 classes. Figure 1 shows ten light curves per class, normalized by their maximum flux and aligned relative to the peak time. Distinct differences can be observed between classes. Supernova-like objects generally exhibit a prominent peak followed by a gradual decline, whereas classes such as KN, EB, RRL, Mira, and MicroLens-Single show more irregular or rapidly varying behavior.

Table 1: Class distribution of the 3000 experimental subset and the full PLAsTiCC dataset.

	Experimental subset		Full dataset	
Class	Count	Proportion (%)	Count	Proportion (%)
AGN	136	4.5	370	4.7
EB	349	11.6	924	11.8
KN	44	1.5	102	1.3
M-dwarf	399	13.3	981	12.5
MicroLens-Single	55	1.8	151	1.9
Mira	10	0.3	30	0.4
RRL	77	2.6	239	3.0
SLSN-I	69	2.3	175	2.2
SNII	450	15.0	1193	15.2
SNIa	874	29.1	2313	29.5
SNIa-91bg	76	2.5	208	2.7
SNIax	69	2.3	183	2.3
SNIbc	191	6.4	484	6.2
TDE	201	6.7	495	6.3
Total	**3000**	**100.0**	**7848**	**100.0**

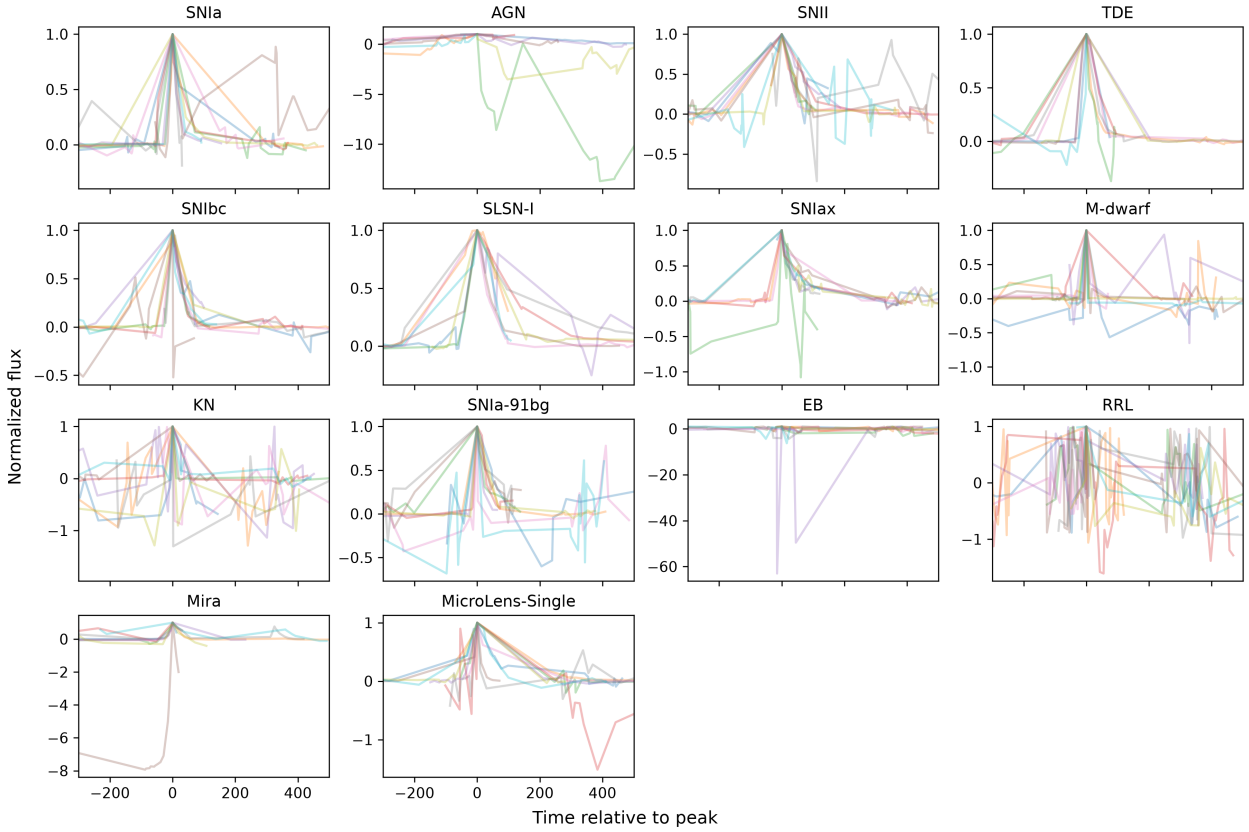


Figure 1: Examples of normalized and peak-aligned r-band light curves across 14 astronomical object classes. Ten objects are shown for each class.

To further investigate class-dependent characteristics across photometric bands, we aggregate normalized flux observations within common time bins and compute the mean light curve for each class and band. As shown in Figure 2, supernova-like classes generally exhibit a well-defined peak and relatively smooth temporal evolution, while classes such as AGN, RRL, Mira, and EB display more irregular or oscillatory patterns.

Although different bands often exhibit peaks at similar times, their relative flux amplitudes can differ substantially. For example, the z-band exhibits a relatively prominent peak in SNIa objects, while other classes show different band-dependent amplitude patterns. Such differences are also observed in TDE, M-dwarf, MicroLens-Single, and KN objects. These observations suggest that photometric bands may share

similar temporal structures without necessarily having comparable flux amplitudes.

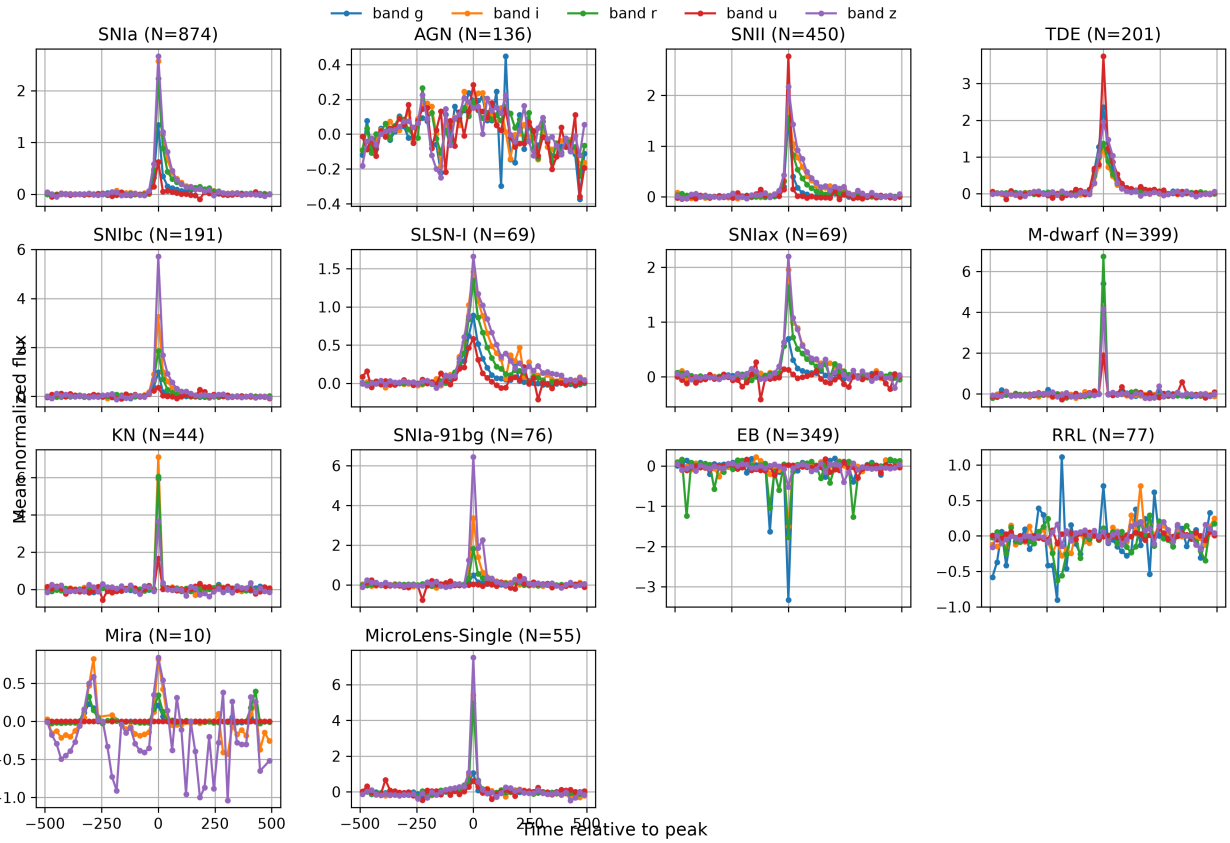


Figure 2: Class-averaged, peak-aligned light curves across five photometric bands for 14 astronomical object classes. Flux values are aggregated within common time bins, and N denotes the number of objects in each class.

However, class-level aggregation has several limitations. The number of objects varies substantially across classes, resulting in different levels of statistical reliability. Furthermore, averaging may obscure object-specific temporal structures and amplitude differences, particularly for classes exhibiting irregular variability. Therefore, class-level patterns should be interpreted as descriptive summaries rather than representative relationships for every individual object.

The similarities between photometric bands suggest that incorporating auxiliary-band observations may improve the reconstruction of sparsely observed target bands. Figure 3 illustrates the relationships between different band pairs for the SNIbc class. Positive correlations are particularly apparent for the g–r, r–i, r–z, and i–z pairs, especially around the peak region, indicating similarities in their temporal evolution. In contrast, pairs involving the u-band exhibit weaker apparent relationships, potentially reflecting differences in flux variability and temporal structure.

These observations motivate the use of wavelength-dependent multi-output Gaussian processes to exploit cross-band correlations. However, temporal similarity alone does not guarantee successful information transfer, as differences in flux amplitude and light-curve morphology may also affect reconstruction accuracy and uncertainty calibration. The influence of these factors is investigated in the subsequent experiments.

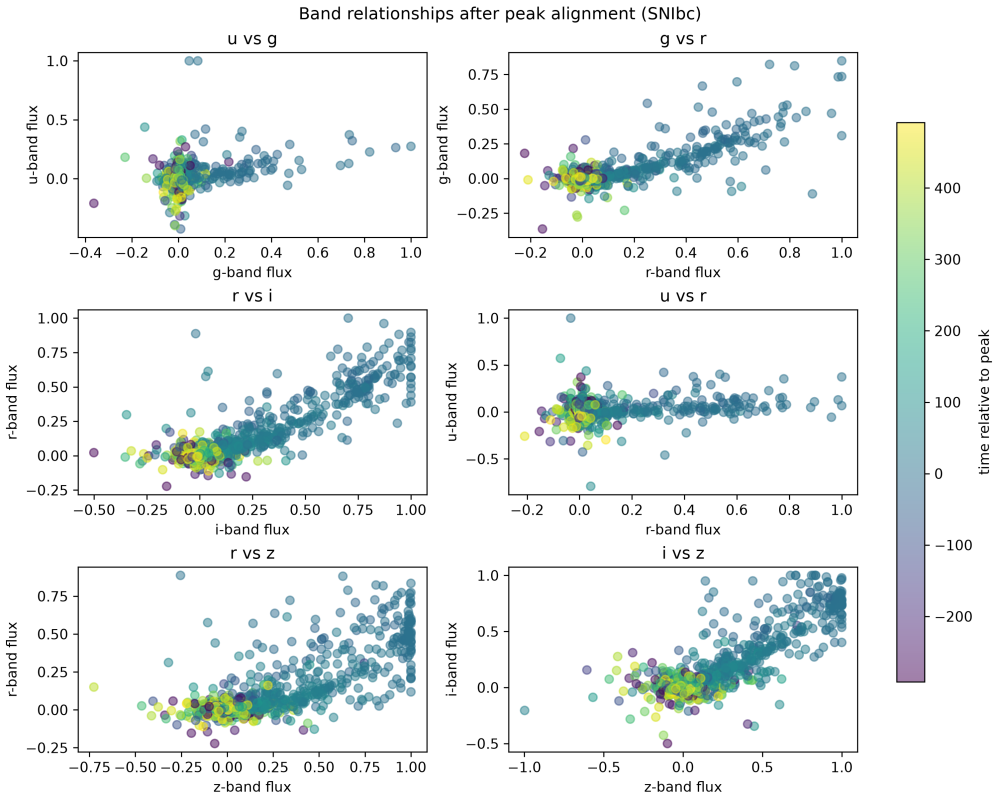


Figure 3: Cross-band flux relationships in peak-aligned SNIbc light curves. Each panel shows the normalized flux values for a pair of photometric bands, with colors indicating the time relative to the peak.

2.3.1 Class-Level Inter-Band Distance Analysis

The wavelength kernel assumes that bands with similar central wavelengths have stronger correlations:

$$|\lambda_b - \lambda_{b'}| \downarrow \quad \Rightarrow \quad k_\lambda(\lambda_b, \lambda_{b'}) \uparrow .$$

However, wavelength proximity does not necessarily imply similar observed light curves. Since correlation coefficients mainly describe similarity in temporal patterns without accounting for amplitude differences, we use an amplitude-sensitive distance to compare light curves across bands.

For each object n, the distance between bands b and b' is defined as

$$D_{n,bb'} = \sqrt{\text{mean}\left[(f_{n,b} - f_{n,b'})^2\right]},$$

where $f_{n,b}$ and $f_{n,b'}$ are the normalized and time-aligned flux values in common bins. Only bins with valid observations from both bands are considered.

The distances are then aggregated within each class using the median:

$$D_{c,bb'} = \text{median}_{n \in c} D_{n,bb'},$$

where c denotes the astronomical object class.

To account for irregular sampling, the following preprocessing steps are applied:

1. Align the observation times of each object relative to its estimated peak.
2. Normalize all bands of each object using a shared flux scale, defined as the median of the five largest absolute flux values across all bands.
3. Divide the relative-time range $[-500, 500]$ into 49 equal-width bins and compute the median flux within each bin.
4. Calculate the root-mean-square difference between each band pair using bins with valid observations in both bands.
5. Aggregate the object-level distances within each class using the median.

This procedure captures differences in both temporal structure and relative flux amplitude while reducing the effects of irregular sampling and extreme observations.

Figure 4 shows the class-level inter-band distance matrices. Smaller distances indicate greater similarity between the normalized light curves. For most supernova-like classes, neighboring bands such as r–i and i–z exhibit relatively small distances, while pairs involving the u-band generally show larger differences. However, the patterns vary across classes. For example, AGN, KN, and RRL exhibit relatively large inter-band distances, while EB shows smaller distances across most band pairs.

These results suggest that wavelength separation is related to empirical cross-band similarity, but the relationship also depends on the object class. Therefore, wavelength proximity alone may not be sufficient to determine the effectiveness of cross-band information transfer.

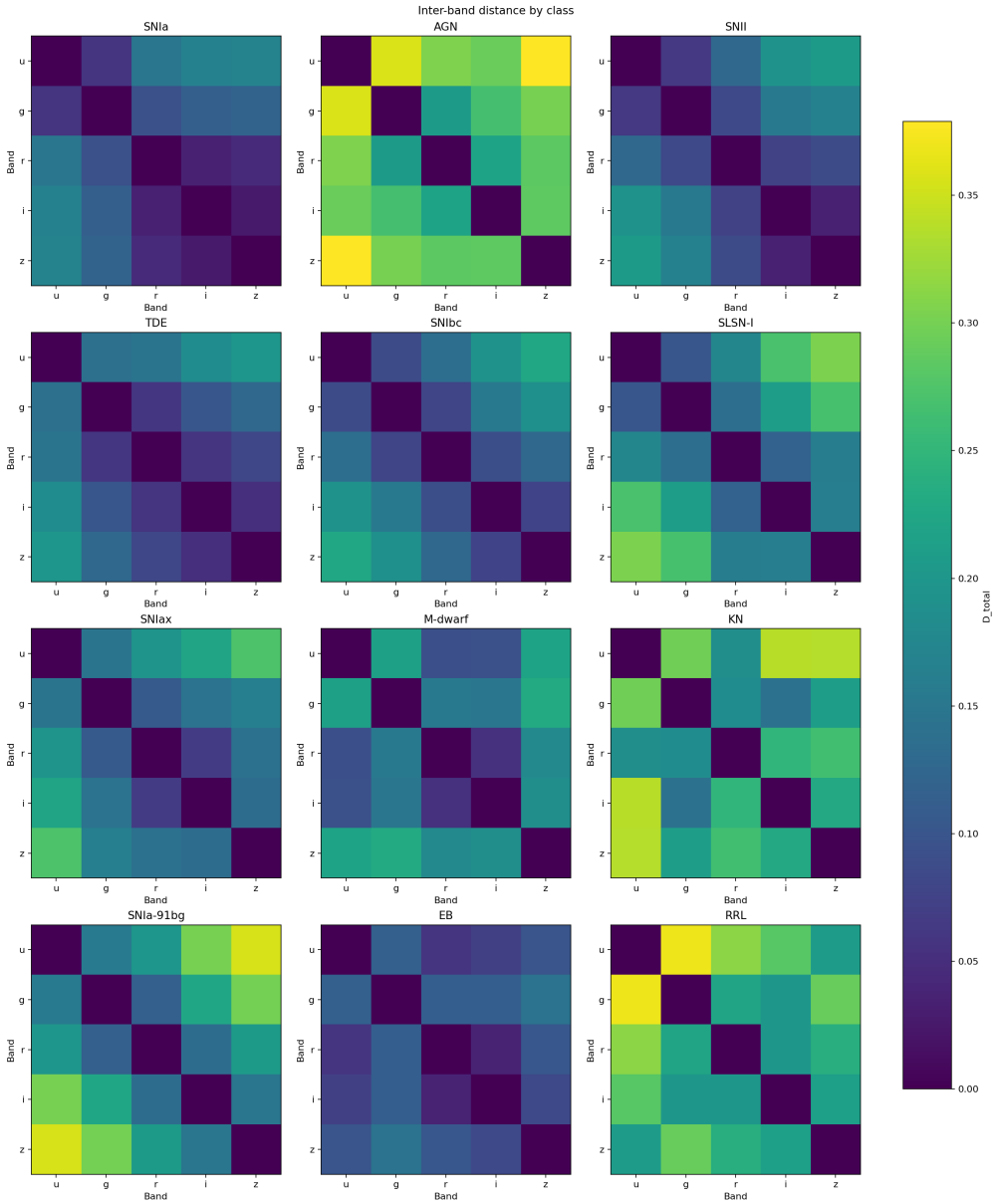


Figure 4: Class-level inter-band distances across five photometric bands. Each matrix shows the median object-level distance between band pairs, with smaller values indicating greater similarity in normalized flux amplitude and temporal evolution.

3 Gaussian-process models

3.1 Gaussian process regression

Gaussian process is a joint Gaussian probability distribution with mean vector $f(t_i)$ and covariance function $k(x, x')$.

$$f(\mathbf{x}) \sim \mathcal{N}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')). \tag{1}$$

Covariance matrix acts as a kernel function that provides the covariance element between any two samples denoted as $\mathbf{x}$ and $\mathbf{y}$, and it can be defined as

$$\mathbf{K}(\mathbf{x}, \mathbf{y}) = \begin{pmatrix} k(x_1, y_1) & k(x_1, y_2) & \cdots & k(x_1, y_n) \\ k(x_2, y_1) & k(x_2, y_2) & \cdots & k(x_2, y_n) \\ \vdots & \vdots & \vdots & \vdots \\ k(x_n, y_1) & k(x_n, y_2) & \cdots & k(x_n, y_n) \end{pmatrix}. \tag{2}$$

Given the time series $[x_1, x_2, ..., x_n]$ and its corresponding observed flux value $[y_1, y_2, ..., y_n]$, we aim at evaluating a set of flux values $\mathbf{y}_*$ at any set of given locations, say $\mathbf{x}_*$

$$p\left(\begin{bmatrix} \mathbf{y} \\ \mathbf{y}_* \end{bmatrix}\right) = \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}(\mathbf{x}) \\ \mu(\mathbf{x}_*) \end{bmatrix}, \begin{bmatrix} \mathbf{K}(\mathbf{x}, \mathbf{x}) & \mathbf{K}(\mathbf{x}, \mathbf{x}_*) \\ \mathbf{K}(\mathbf{x}_*, \mathbf{x}) & k(\mathbf{x}_*, \mathbf{x}_*) \end{bmatrix}\right), \tag{3}$$

The posterior distribution over $\mathbf{y}_*$ can be expressed as

$$P(\mathbf{y}_* | \mathbf{x}_*, \mathbf{x}, \mathbf{y}) = \mathcal{N}(\mu_*, \mathbf{K}_*) \tag{4}$$

where

$$\mu_* = \mu(\mathbf{x}_*) + \mathbf{K}(\mathbf{x}_*, \mathbf{x})\mathbf{K}(\mathbf{x}, \mathbf{x})^{-1} (\mathbf{y}(\mathbf{x}) - \mu(\mathbf{x})) \tag{5}$$

and

$$\mathbf{K}_* = \mathbf{K}(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{K}(\mathbf{x}_*, \mathbf{x})\mathbf{K}(\mathbf{x}, \mathbf{x})^{-1}\mathbf{K}(\mathbf{x}_*, \mathbf{x})^T \tag{6}$$

The covariance function $k(x, x')$ is predefined through the choice of kernel. Astronomical light curves can exhibit both gradual evolution and relatively rapid temporal variations. We therefore use the Matérn 3/2 kernel, which provides a moderately smooth prior without imposing the strong smoothness assumption of the squared-exponential kernel. The kernel is stationary and depends only on the distance $|x - x'|$ between two inputs.

$$k(x, x') = \sigma_f^2 \left(1 + \frac{\sqrt{3}|x - x'|}{\ell}\right) \exp\left(-\frac{\sqrt{3}|x - x'|}{\ell}\right), \tag{7}$$

3.2 Single-band Gaussian process

In this experiment, GP only takes in the flux of a single band. From multiple experiment results, The use of global and local scale, as well as the background reduction, do not have a recognizable effect on single-band GP. So we opt to use local scale and background reduction. The curves can be scaled back to its original scales for downstream tasks like classification.

To further investigate the single-band GP's performance, instead of grouping objects only by class, we compute object-level or object-band-level features from several categories: morphology, fluctuation or roughness, signal-to-noise ratio, and sampling geometry. Then GP performance is evaluated at the object-band level and analyze how the evaluation metrics vary with these features.

3.3 Wavelength-dependent Multi-output Gaussian Process

The multi-output Gaussian process (MOGP) jointly models light curves from different photometric bands by treating wavelength as an additional input dimension. Although the model predicts a scalar flux value for each time–wavelength pair, it can equivalently be interpreted as a multi-output GP, where each photometric band represents a distinct but correlated output function. Cross-band dependencies are modeled using a separable time–wavelength covariance kernel, following [5].

$$k\left((t, \lambda), (t', \lambda')\right) = C \cdot k_{\text{time}}(t, t') \cdot k_{\text{wavelength}}(\lambda, \lambda'), \tag{8}$$

where t denotes the observation time, λ is the central wavelength of the photometric band, and $C > 0$ is a shared covariance amplitude. Both the temporal and wavelength kernels use the Matérn-3/2 covariance function, with the wavelength length scale fixed at 6000 Å.

The shared amplitude C assumes the same marginal prior variance across all photometric bands, while the wavelength kernel controls the strength of cross-band covariance. This allows observations from auxiliary bands to contribute to target-band reconstruction. However, the shared-amplitude assumption may be restrictive when different bands exhibit substantially different flux amplitudes, which we will discuss in Section 5.2.3.

3.4 Train–test construction and preprocessing

Before comparing single- and multi-output GPs, we performed preliminary sensitivity experiments on train-test construction, normalization, and background treatment. These experiments showed that calibration was strongly affected by whether the peak region was represented in the training data, whereas reasonable normalization and background-subtraction choices had comparatively smaller effects. The preprocessing pipeline was therefore fixed before the subsequent model comparisons.

Specifically, preliminary split-sensitivity experiments showed that GP performance deteriorates substantially when the peak is absent from the training set, indicating that calibration depends strongly on the temporal support of the observations. We therefore preserve the maximum flux value in the single band GP experiment and in the fraction 0.8 target training subset in MOGP experiment.

All bands share one reference time, which reference is the timestamp of the highest-flux observation in the selected target-band training subset. Since the kernels are stationary, this translation does not affect their covariance values.

All normalization parameters are estimated exclusively from the training observations to avoid leakage from the held-out test points. Preliminary comparisons between global and local scaling showed only modest differences in aggregate calibration, so a fixed training-only normalization scheme was used for all subsequent experiments.

Preliminary preprocessing experiments also considered robust background subtraction, but its effect on aggregate GP calibration was negligible; it was therefore not pursued further.

4 Evaluation of Predictive Uncertainty

4.1 Point prediction accuracy

Root mean squared error (RMSE) The common expression for RMSE is

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \mu_i)^2}. \tag{9}$$

The choice of N matters. Object-weighted RMSE is weighted by the number of objects, whereas observation-weighted RMSE is weighted by the number of observations. Nevertheless, raw RMSE is hard to compare across objects, since PLAsTiCC objects have very different flux scales. Therefore, a normalized RMSE (NRMSE) would help in making RMSE comparable across objects. The formula below is adopted

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{n}\sum_i (y_i^{test} - \hat{y}_i)^2}}{s} \tag{10}$$

where

$$s = max\left(P_{95}\left(|y_{all}|\right), \epsilon\right) \tag{11}$$

It normalizes RMSE using the maximum of 95th percentile of flux value from that object.

Note that the scale is only used for post-hoc evaluation, not for training, hyperparameter optimization, model fitting, or prediction. Hence it uses an all-observed-data object-level empirical flux scale.

4.2 Probabilistic scoring rules

Negative log predictive density (NLDP) For each test point i, GP gives a predictive distribution $p(Y_i|\mathbf{x_i}) \sim \mathcal{N}(\mu_i, \sigma_i^2)$. Given the observed y_i, the predictive density assigned to the observation is

$$p(y_i|\mathbf{x_i}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left[-\frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right] \tag{12}$$

Take the negative logarithm and average it over N observations, it gives the formula for NLPD

$$\text{NLPD} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{2}\log(2\pi\sigma_i^2) + \frac{(y_i-\mu_i)^2}{2\sigma_i^2}\right]. \tag{13}$$

The first term penalizes unnecessarily large predictive uncertainty, while the second penalizes prediction errors relative to the predicted variance. Consequently, NLPD rewards predictive distributions that assign high probability density to the observations, while penalizing both overly diffuse predictions and overconfident incorrect predictions. Lower NLPD indicates better probabilistic predictive performance.

Continuous ranked probability score (CRPS) RMSE evaluates the predictive mean, while coverage and PIT assess aspects of calibration but do not directly reward sharp predictive distributions. CRPS instead evaluates the entire predictive distribution by comparing the predicted cumulative distribution function (CDF) and the observed step CDF at all possible thresholds, then summing up the squared difference. It therefore accounts jointly for the accuracy and spread of the predictive distribution, rewarding distributions that are concentrated while remaining consistent with the observations. The CRPS loss is usually written as

$$\begin{aligned}\text{CRPS}(F, y) &= \int_{-\infty}^{\infty}(F(x) - \mathbf{1}\{x \geq y\})^2 dx, \\ &= \mathbb{E}_F|X - y| - \frac{1}{2}\mathbb{E}_F|X - X'|,\end{aligned} \tag{14}$$

where $y \in \mathbb{R}$, X, and X' are independent random variables with predicted CDF) F of X. $\mathbb{I}$ is the Heaviside step function, which yields 1 if the real argument is positive or zero, otherwise 0. Mathematically, CRPS is the integrated squared distance between the predicted CDF and the CDF corresponding to the observation. The expectation representation illustrates how CRPS balances the distance of the predictive distribution from the observation against the variance of the predictive distribution [10]. For Gaussian distribution, CRPS can be written as

$$\begin{aligned}\text{CRPS}(\mathcal{N}(\mu, \sigma^2), x) = \sigma\Bigg[&\frac{1}{\sqrt{\pi}} - 2\varphi\left(\frac{x-\mu}{\sigma}\right) \\ &- \frac{x-\mu}{\sigma}\left(2\Phi\left(\frac{x-\mu}{\sigma}\right) - 1\right)\Bigg],\end{aligned} \tag{15}$$

where φ and Φ are the probability density function and the CDF of a standard Gaussian variable [11].

CRPS is more robust compared with NLPD, especially when GP is severely overconfident on a few points, while CRPS is generally less dominated by such extreme density errors. Therefore, we mainly use CRPS for evaluation in this experiment.

4.3 Calibration diagnostics

Standardized residuals. The definition of standardized residual is

$$z_i = \frac{y_i - \mu_i}{\sigma_i}. \tag{16}$$

where y_i stands for the observed flux value at measured time i, and μ_i and σ_i are the mean and standard deviation of the distribution we get from GP. It measures the residual of the observed flux and the predicted flux. For a calibrated Gaussian predictive distribution, $mean(z) \sim 0$ and $std(z) \sim 1$. If a distribution is underconfident, $std(z) < 1$. If a model accurately predicts the data points so that $|y_i - \mu_i|$ is relatively small, whereas the standard deviation σ_i assigned to these points is relatively large. Then the standard deviation of all z_i values drops below 1 because almost all observed values y_i fall tightly inside the 1σ interval.

Prediction interval coverage. Prediction interval coverage measures the proportion of observed values that fall within the predicted uncertainty intervals. For a Gaussian predictive distribution, the empirical coverage at k standard deviations is defined as

$$\text{Coverage}_{k\sigma} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(|y_i - \mu_i| \leq k\sigma_i\right), \tag{17}$$

where $\mathbb{I}$ is the indicator function, which equals 1 if the observed value falls within the interval and 0 otherwise.

For a well-calibrated Gaussian predictive distribution, the nominal coverage levels at 1σ, 2σ, and 3σ are approximately 68.3%, 95.4%, and 99.7%, respectively. If the empirical coverage is higher than the nominal level, the predictive intervals are too wide, indicating underconfidence. Conversely, lower empirical coverage suggests that the intervals are too narrow, indicating overconfidence. By comparing empirical and nominal coverage, we can evaluate whether the GP provides reliable uncertainty estimates.

Probability integral transform (PIT) PIT is a calibration diagnostic for probabilistic forecasts. Given a cumulative distribution function (CDF) F and observed value y, the PIT value is $z = F(y)$. According to probability integral transform, for a well-calibrated probabilistic model, the PIT values should approximately follow a uniform distribution on $[0, 1]$, and the empirical PIT reliability curve should lie close to the diagonal. A concentration of PIT values around 0.5 indicates that the predictive distributions are too wide, i.e. the model is underconfident. In contrast, a U-shaped PIT histogram, with too many PIT values close to 0 and 1, indicates that the predictive distributions are too narrow, i.e. the model is overconfident.

We will carry out the stratified PIT analysis, where the objects are splited into bins by training set size(e.g., sparse/medium/dense tertiles), class, and band. If the model is calibrated, each bin should contain approximately the same fraction of points.

KS-PIT Kolmogorov-Smirnov (KS) distance computes the maximum absolute distance between two CDF. To obtain a quantified measurement of the PIT, we can apply KS distance to PIT values [12], which yields the KS-PIT metric

$$KS - PIT = sup \left| \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}\{U_i \leq u\} - u \right| \tag{18}$$

The value of the KS distance falls between 0 (identical distributions) and 1 If KS-PIT is 0, it implies identical distributions, whereas 1 value means the distributions are completely non-overlapping.

5 Experiments and Results

5.1 Single-band GP calibration

The target band is set to 'r', the flux of which has relatively large magnitude. To filter out light curve with too few data points, the minimum number of data points required to keep the example is 8. The scale for flux normalization is computed locally within each light curve.

5.1.1 Overall and class-dependent calibration

Classes' distinct features correspond to various GP calibration performance. We apply GP on each object and aggregate the results per class. The result is shown in Table 2.

Table 2: Class-wise predictive performance and calibration of the single-band GP. N denotes the number of light curves in each class, $\bar{n}_{\text{train}}$ is the mean number of target-band training observations, and s_{flux} is the mean object-level flux scale used for normalization. Coverage denotes the empirical 1σ predictive interval coverage. NRMSE and NCRPS are normalized by the object-level flux scale.

Class	N	$\bar{n}_{\text{train}}$	s_{flux}	NRMSE	NCRPS	NLPD	$\text{Cov.}_{1\sigma}$	$\bar{z}$	σ_z	KS-PIT
SNIa	872	23.77	5.91	2.662	1.694	3.461	0.803	-0.080	0.728	0.366
SNII	450	21.56	6.73	2.612	1.578	3.643	0.782	-0.085	0.792	0.372
M-dwarf	399	23.40	18.45	5.817	4.439	7.356	0.793	-0.183	1.127	0.391
EB	348	20.92	147.43	6.864	4.791	9.278	0.739	0.075	1.352	0.399
TDE	200	16.86	9.05	5.202	3.121	3.733	0.814	-0.107	0.682	0.398
SNIbc	191	21.99	6.10	2.310	1.600	3.439	0.797	-0.088	0.716	0.374
AGN	136	23.35	121.58	0.294	0.170	4.128	0.621	-0.036	1.082	0.358
SNIa-91bg	76	23.24	5.70	1.506	0.981	3.120	0.801	-0.034	0.676	0.367
RRL	76	22.18	1811.31	2.140	1.632	13.412	0.635	-0.070	1.919	0.388
SLSN-I	69	21.68	16.59	0.841	0.650	3.328	0.771	0.018	0.745	0.372
SNIax	69	25.72	6.25	1.478	0.993	3.238	0.785	-0.060	0.772	0.347
MicroLens-Single	55	16.25	1794.59	2.220	19.081	4.851	0.768	-0.122	0.635	0.401
KN	43	17.74	5.54	1.744	2.208	4.000	0.843	-0.277	0.788	0.373
Mira	10	25.10	3881.97	0.093	0.126	6.536	0.888	-0.009	0.477	0.404
Overall	2994	22.19	121.51	3.461	2.684	5.010	0.778	-0.076	0.904	0.377

From the experiment on 3000 objects, the single-band GP shows clear class- dependent behavior.

EB and RRL are problematic classes. the mean RMSE is very large, about 420, and the normalized RMSE is also extremely large, about 9.72. That means the GP is doing badly relative to the object scale. Calibration is also poor: $z_std_mean \approx 1.45$, 2σ coverage is only 0.88, and 3σ coverage is 0.925, both below the Gaussian

ideal values of about 0.954 and 0.997. So for EB, the model is both inaccurate and overconfident. RRL only has 5 objects so we need more samples before making a firm class-level claim.

For the supernova-like classes, the GP is more stable. They mostly have much lower RMSE/NLPD/CRPS than EB/RRL. But their calibration is often in the opposite direction: many have $z_std_mean < 1$. It implies the model is underconfident.

The situation is mixed for AGN and M-dwarf. They have one of the better z-score standard deviations, suggesting that the average uncertainty scale is not badly wrong. However, their empirical coverage is below the nominal Gaussian levels, so the predictive distribution is still not fully calibrated and appears mildly overconfident or heavy-tailed. We need PIT histogram for further examination.

Because Mira, KN, and MicroLens-Single have the less number of light curves, we only consider other classes' light curves in the following experiment.

As seen in Figure 5, for the supernova-like classes, including SLSN- I, SNII, SNIa, SNIa-91bg, SNIbc, SNIax, and TDE, PIT histograms show a clear concentration around 0.5 and a depletion near both tails. PIT reliability curve lies below the diagonal for small probabilities and above the diagonal for large probabilities, forming an S-shaped deviation. This indicates that the predictive distributions are underconfident: the held-out observations fall too often near the centre of the predictive CDF. This observation is consistent with $z_{\text{std}} < 1$. The small asymmetry of the histogram and reliability curve also suggests that a weak bias component may be present.

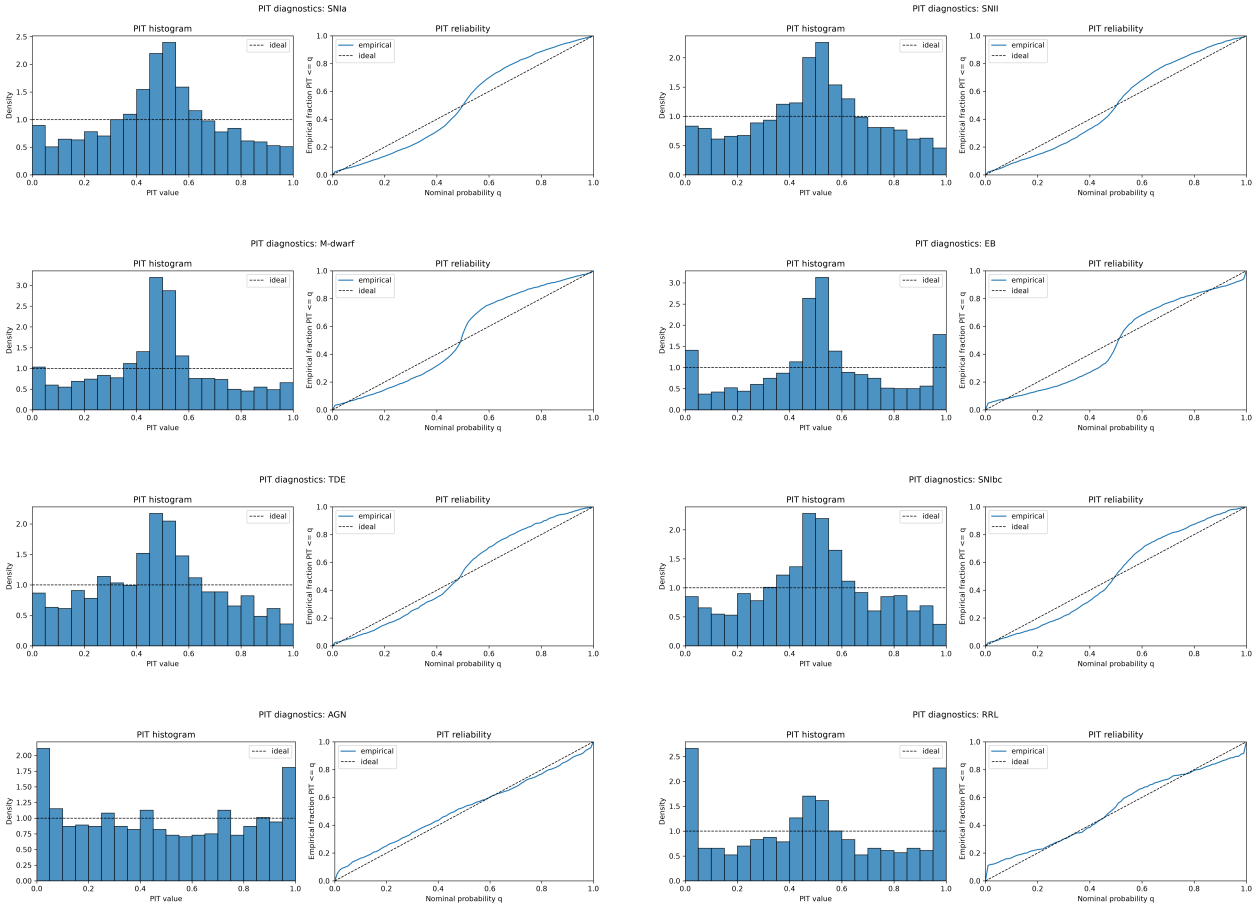


Figure 5: PIT diagnostics for different object classes. Each panel shows the PIT histogram (left) and empirical PIT reliability curve (right).

5.1.2 Prediction error versus predictive uncertainty

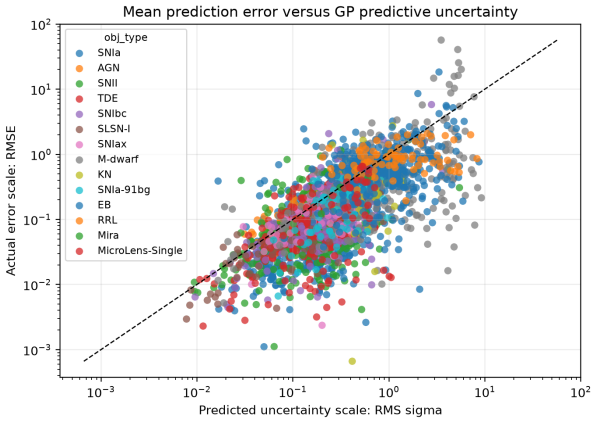
In this experiment, each object-band pair is represented as one point in a scatter plot. The horizontal axis shows the root-mean-square predictive uncertainty over the held-out observations, while the vertical axis shows the root-mean-square error (RMSE) of the predictive mean:

$$\text{Predictive uncertainty scale} = \sqrt{\text{mean}\left(\sigma^2_{\text{pred}}\right)}$$

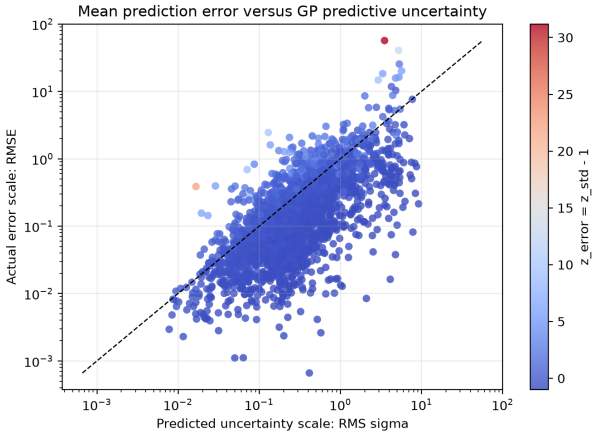
$$\mathrm{RMSE} = \sqrt{\mathrm{mean}\left(y_{\mathrm{true}} - \mu_{\mathrm{pred}}\right)^2}$$

A diagonal reference line, $y = x$, is added to the plot. Points close to this line have prediction errors whose overall magnitude is comparable to the predicted uncertainty. Points above the line have errors larger than their predicted uncertainties and therefore indicate an overconfident tendency. Points below the line have uncertainties larger than their errors and indicate an underconfident tendency.

The results are shown in Figures 6a and 6b.



(a) Object-band points labelled by transient class.



(b) Object-band points coloured by the standardized error metric z_{error}.

Figure 6: Comparison between the predictive uncertainty scale and the RMSE of the predictive mean.

Most points lie below the diagonal and are associated with relatively small standardized errors. This indicates that many of the fitted GPs are underconfident: their predictive uncertainties are larger than the observed prediction errors.

The extreme red point in Figure 6b represents a pathological overconfident case. The corresponding light curve is shown in Figure 7. The GP fails to reconstruct a sharp peak that is supported by only one high-flux observation. Although this observation is included in the training set, a single point is insufficient to constrain the width and shape of the peak. The GP consequently produces a smooth prediction that misses the held-out measurements near the peak while assigning insufficient uncertainty to this failure.

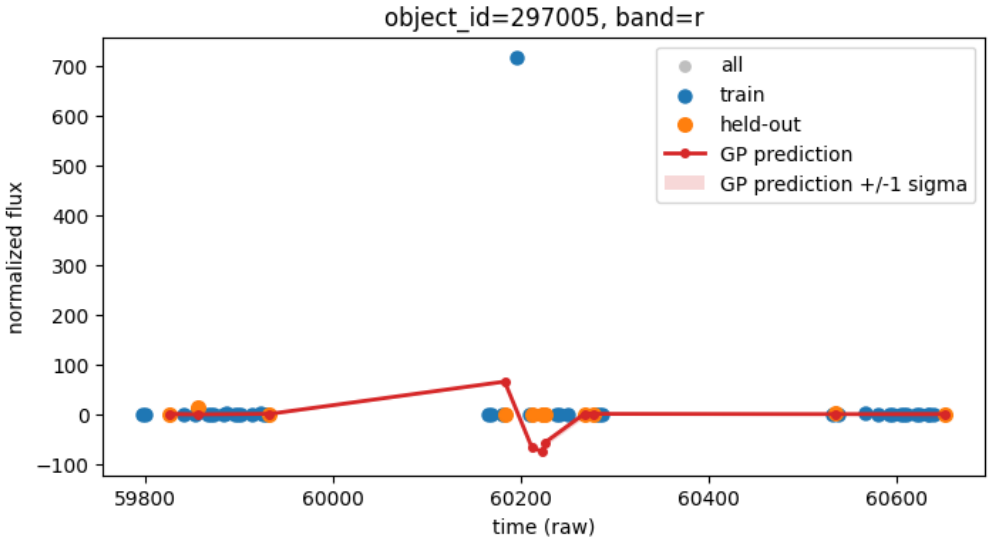


Figure 7: Predicted light curve for the object corresponding to the extreme red point in Figure 6b.

5.2 Cross-band information transfer in the MOGP

5.2.1 Dependence on data sparsity

To investigate how target-band data availability affects the benefit of cross-band information, we compare three models: single-band GP, independent-band GP, and real-wavelength multi-band GP. The independent-band GP uses the same training observations as the real-wavelength GP but suppresses cross-band covariance through a very small wavelength length scale. However, it still estimates shared kernel hyperparameters using all input bands, allowing us to distinguish the effects associated with shared hyperparameter estimation and wavelength-dependent modeling.

We use the i band as the target and retain 50% of the observations from each auxiliary band (u, g, r, z). A fixed 20% of target-band observations is reserved for evaluation, while the remaining 80% forms the training pool. Nested training subsets are constructed with target fractions ranging from 0.1 to 0.8. The experiment includes 300 objects, except at fraction 0.1, where 286 objects satisfy the minimum sampling requirements.

To account for differences in flux amplitude across objects, we use normalized CRPS (nCRPS) and nRMSE as the primary performance metrics. Both metrics are normalized by the same object-level flux scale:

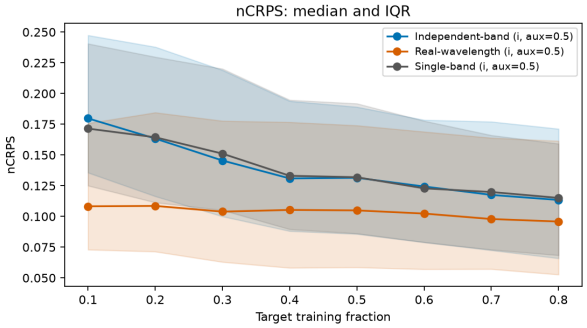
$$S_o = \max\left(Q_{95}\left(\left\{|F_b(t)| : b \in \mathcal{B},\, t \in \mathcal{T}_b^{\text{valid}}\right\}\right), 10^{-8}\right), \tag{19}$$

where $\mathcal{B}$ contains all available photometric bands, and the quantile is calculated from the complete valid observations of each object. The normalized metrics are defined as

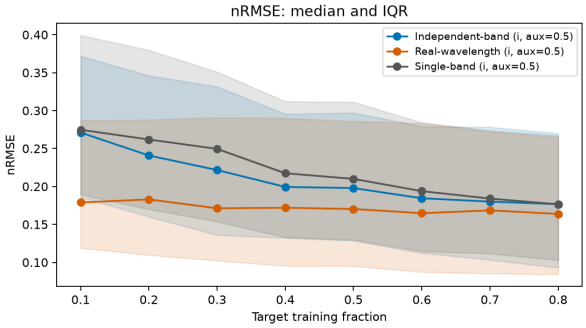
$$\text{nCRPS} = \frac{\text{CRPS}}{S_o}, \qquad \text{nRMSE} = \frac{\text{RMSE}}{S_o}. \tag{20}$$

The same denominator is used for all models, target fractions, and random splits of an object. These evaluation metrics are distinct from the normalization applied internally during GP fitting.

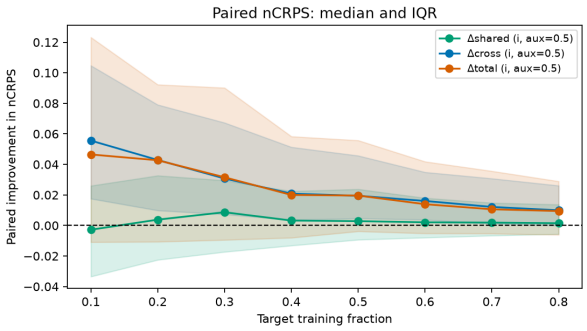
Figure 8 shows that the real-wavelength GP achieves the lowest median nCRPS and nRMSE across all target fractions. Its advantage is particularly pronounced when target-band observations are sparse. At a target fraction of 0.1, median nCRPS decreases from 0.171 (single-band) and 0.180 (independent-band) to 0.108 (real-wavelength), while median nRMSE decreases from approximately 0.275 and 0.271 to 0.179. At fraction 0.8, the corresponding median nCRPS values converge to 0.115, 0.113, and 0.096, respectively.



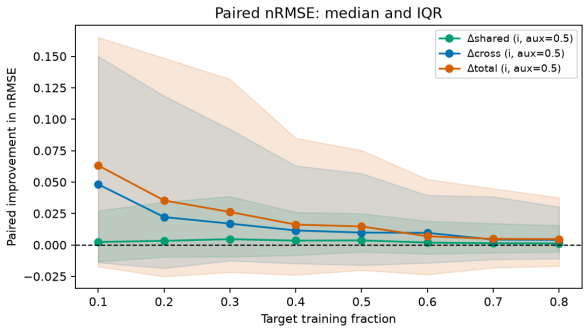
(a) Median nCRPS and IQR.



(b) Median nRMSE and IQR.



(c) Paired improvement in nCRPS.



(d) Paired improvement in nRMSE.

Figure 8: Dependence of GP reconstruction performance on target-band training fraction, with the auxiliary-band fraction fixed at 0.5. Solid lines indicate median performance, and shaded regions represent the IQR. Positive paired improvements indicate better performance of the latter model in each comparison.

Table 3: Median CRPS, nCRPS, and nRMSE across target-band training fractions. Raw CRPS is reported in original flux units, while nCRPS and nRMSE are dimensionless.

Fraction	Model	CRPS	nCRPS	nRMSE
0.1	Single-band	8.634	0.171	0.275
	Independent	9.159	0.180	0.271
	Real-wavelength	5.684	0.108	0.179
0.2	Single-band	8.430	0.164	0.262
	Independent	8.388	0.163	0.241
	Real-wavelength	5.615	0.109	0.183
0.4	Single-band	7.119	0.133	0.217
	Independent	7.230	0.131	0.199
	Real-wavelength	5.343	0.105	0.172
0.8	Single-band	5.431	0.115	0.177
	Independent	5.704	0.113	0.177
	Real-wavelength	4.681	0.096	0.164

Another notable observation is that median nCRPS remains relatively stable across target fractions, decreasing only from 0.108 to 0.096. In contrast, the single-band and independent-band models show much larger improvements as more target observations become available, accompanied by decreasing inter-object performance variability. This suggests that cross-band information may reduce the dependence of probabilistic reconstruction performance on target-band sampling density.

To quantify the improvement for individual objects, we define three paired differences:

$$\begin{aligned} \Delta_{\text{shared}} &= M_{\text{single}} - M_{\text{ind}}, \\ \Delta_{\text{cross}} &= M_{\text{ind}} - M_{\text{real}}, \\ \Delta_{\text{total}} &= M_{\text{single}} - M_{\text{real}}, \end{aligned} \tag{21}$$

where M denotes either nCRPS or nRMSE, and positive differences indicate an improvement.

The paired results in Figure 8 show that shared hyperparameter estimation alone provides relatively small and inconsistent median improvements. In contrast, the real-wavelength GP provides substantial additional gains, particularly under sparse target observations. Median paired Δ_{cross} in nCRPS decreases from 0.0556 at fraction 0.1 to 0.0099 at fraction 0.8. Similarly, the paired nRMSE improvements decrease as target-band observations become more abundant. These trends indicate diminishing benefits from additional cross-band information as the target band becomes better constrained.

However, these improvements are not universal, as shown in Figure 9. For nCRPS, the fraction of objects with negative Δ_{cross} increases from approximately 15.4% to 25.3% as the target training fraction increases from 0.1 to 0.8. In contrast, the corresponding fraction for nRMSE is substantially higher, ranging from approximately 33% to 41%. This suggests that wavelength-dependent modeling improves probabilistic prediction quality more consistently than point prediction accuracy.

Interestingly, the fraction of objects with negative Δ_{total} is consistently higher than that with negative Δ_{cross} for nCRPS. This indicates that although the real-wavelength GP improves upon the independent-band control for most objects, it does not always outperform the single-band GP. One possible explanation is that shared hyperparameter estimation across bands can be suboptimal for certain target-band light curves, particularly when different bands exhibit substantially different amplitudes or temporal characteristics. Cross-band covariance may compensate for some of these limitations, but not necessarily enough to outperform a target-only GP.

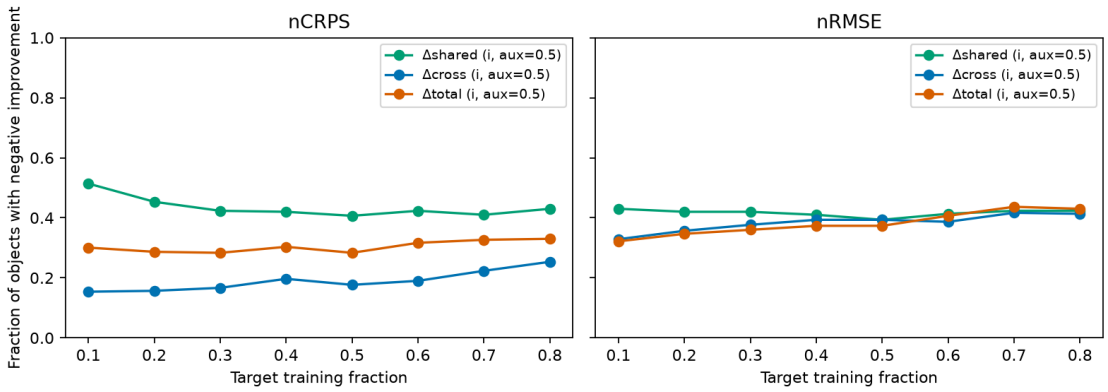


Figure 9: Fraction of objects exhibiting negative paired improvements in nCRPS and nRMSE across target-band training fractions. The three comparisons are Δ_{shared} (single-band versus independent-band), Δ_{cross} (independent-band versus real-wavelength), and Δ_{total} (single-band versus real-wavelength). Negative improvement indicates that the latter model performs worse than the former.

Table 4: Probabilistic calibration across target training fractions. Values are observation-weighted summaries. Ideal standardized residuals have mean zero and standard deviation one; nominal 1σ coverage is 0.683.

Fraction	Model	KS-PIT	z mean	z std	Coverage 1σ
0.1	Single-band	0.436	0.980	4.375	0.652
	Independent	0.422	0.207	1.481	0.812
	Real-wavelength	0.382	0.042	1.425	0.769
0.2	Single-band	0.419	0.631	3.821	0.688
	Independent	0.411	0.136	1.216	0.816
	Real-wavelength	0.383	0.033	1.212	0.774
0.4	Single-band	0.393	0.009	1.421	0.743
	Independent	0.390	-0.001	0.960	0.814
	Real-wavelength	0.376	-0.015	1.004	0.776
0.8	Single-band	0.378	-0.103	0.935	0.779
	Independent	0.378	-0.072	0.794	0.807
	Real-wavelength	0.367	-0.047	0.856	0.776

Table 4 shows that the real-wavelength GP also achieves the lowest KS-PIT values across all sampling fractions, suggesting improved probabilistic calibration. When target observations are sparse, the single-band GP exhibits particularly large standardized residual means and standard deviations. The two multi-band models show considerably smaller residual deviations.

Nevertheless, all three models exhibit imperfect calibration. In particular, both multi-band GPs exhibit persistent overcoverage at the nominal 1σ level, suggesting generally underconfident predictive distributions. The real-wavelength GP shows less overcoverage than the independent-band control, indicating improved interval calibration. However, standardized residuals and KS-PIT statistics reveal that neither model is perfectly calibrated.

Overall, the results demonstrate that wavelength-dependent cross-band modeling is most beneficial when target observations are scarce. Shared hyperparameter estimation alone provides limited median improvement, whereas cross-band covariance enables substantial reconstruction gains, although negative transfer remains possible for individual objects. We will further investigate the failure cases in the following section.

5.2.2 Dependence on light-curve properties

To further investigate the factors associated with GP predictive performance, we examine three explanatory properties: target-band sampling gaps, light-curve roughness, and cross-band temporal similarity. We focus on target training fractions of 0.2 and 0.8, representing sparse and relatively dense target observations, respectively.

Sampling geometry and light-curve morphology. We first examine the relationship between predictive performance and two target-band features: `gap_90_train`, which characterizes temporal gaps in the training observations, and `mean_abs_slope`, which measures light-curve variability. Objects are divided into three approximately equal-sized quantile groups for each feature.

As shown in Figure 10, target-band sampling gaps exhibit no clear monotonic relationship with CRPS or nRMSE. This is consistent with the weak Spearman correlations between `gap_90_train` and CRPS ($|\rho| < 0.10$ across the three models and both target fractions). However, at a target fraction of 0.8, larger gaps are weakly associated with higher KS-PIT values for the single-band and independent-band models ($\rho = 0.191$ and 0.185, respectively), suggesting a possible relationship with poorer probabilistic calibration.

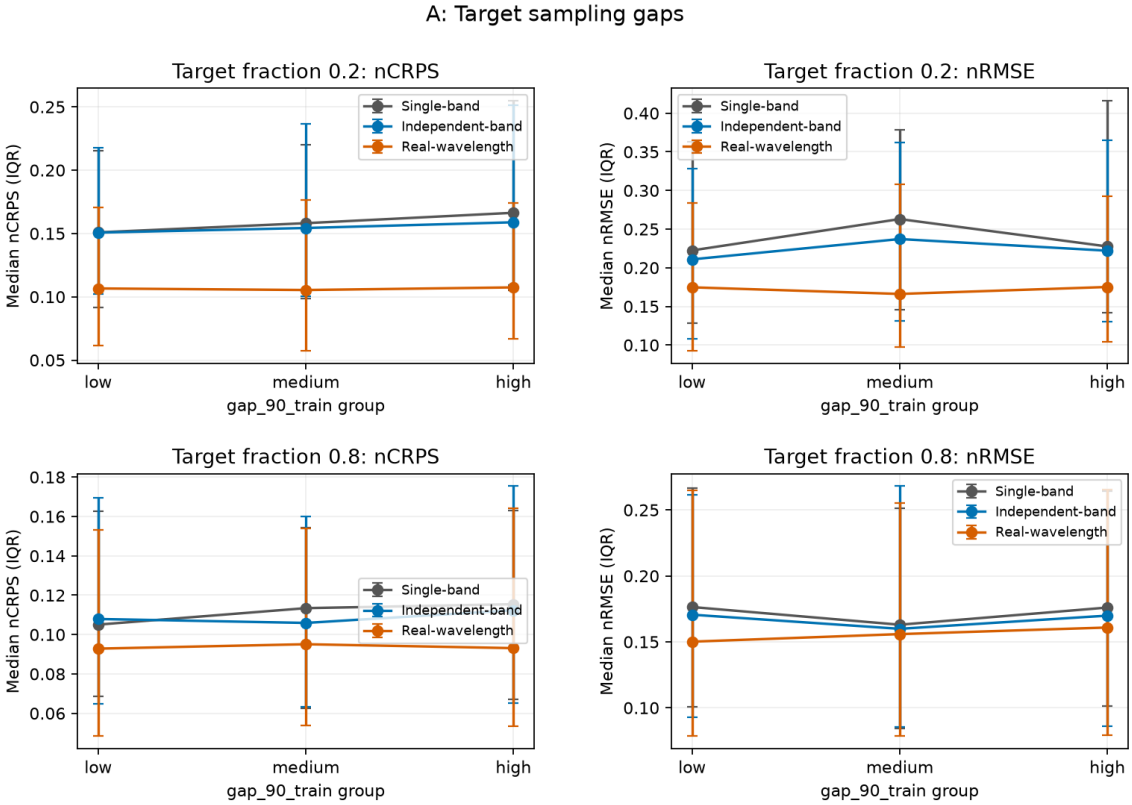


Figure 10: Dependence of GP performance on target-band sampling gaps. The `gap_90_train` feature is defined as the 90th percentile of the time intervals between consecutive target-band training observations. Objects are divided into three quantile-based groups (low, medium, and high). Points indicate median performance, and error bars represent IQR.

In contrast, light-curve roughness shows a strong positive association with raw CRPS and NLPD. At a target training fraction of 0.2, the Spearman correlations between `mean_abs_slope` and raw CRPS are 0.647, 0.578, and 0.517 for the single-band, independent-band, and real-wavelength GPs, respectively. However, this relationship changes after flux normalization.

For nCRPS, the corresponding correlations become 0.089, -0.085, and -0.254, respectively. At a target fraction of 0.8, they are -0.002, -0.076, and -0.128. Similar weak negative associations are observed for nRMSE. As illustrated in Figure 11, the real-wavelength GP generally maintains lower nCRPS and nRMSE across different roughness groups.

This discrepancy between raw CRPS and normalized metrics suggests that the previously observed positive association was partly driven by differences in flux scale, since `mean_abs_slope` is itself calculated from unnormalized flux values. Therefore, the results do not provide evidence that greater temporal variability necessarily makes relative reconstruction more difficult. Instead, the relationship between light-curve morphology and predictive performance depends on the evaluation scale and may also reflect other object-level characteristics.

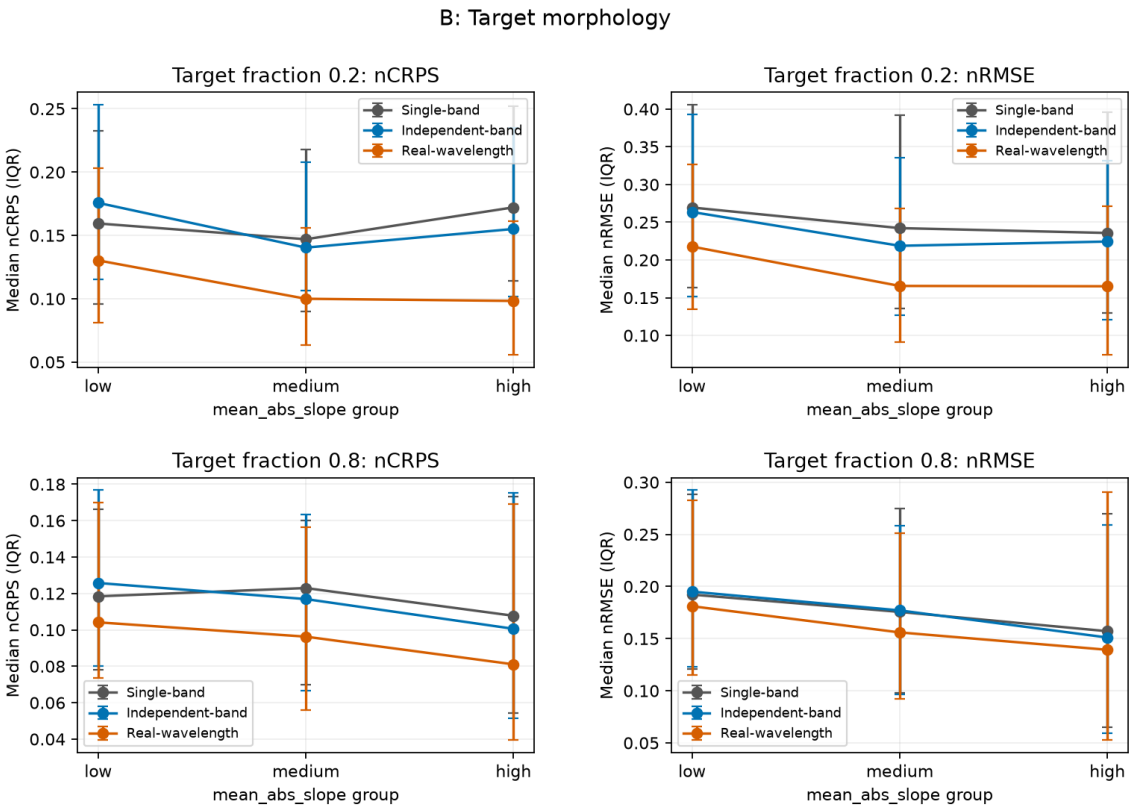


Figure 11: Dependence of normalized GP reconstruction performance on target-band light-curve roughness. The mean absolute slope is computed from consecutive target-band training observations: $\text{mean_abs_slope} = \frac{1}{N-1}\sum_{k=1}^{N-1}\left|\frac{f_{k+1}-f_k}{t_{k+1}-t_k}\right|$. Objects are divided into low, medium, and high quantile groups. Points represent median nCRPS and nRMSE, with error bars indicating the interquartile range (IQR).

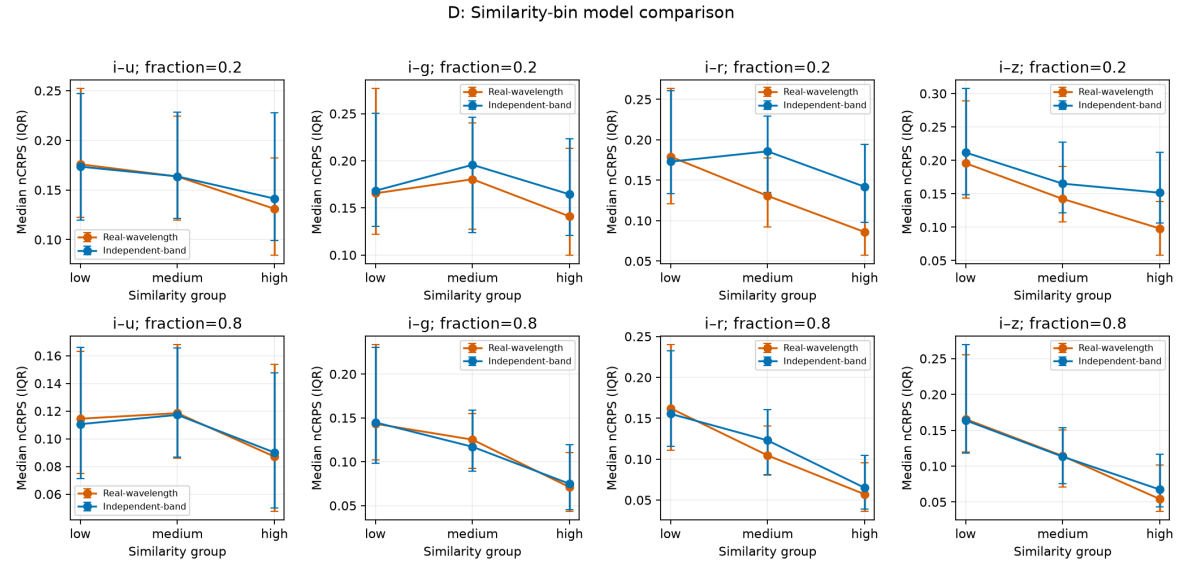
Cross-band temporal similarity. To examine the role of cross-band relationships, we conduct a separate experiment using one auxiliary band at a time. The target band is fixed to i, with target training fractions of 0.2 and 0.8 and an auxiliary training fraction of 0.5.

For each object and target–auxiliary pair, empirical temporal similarity is estimated retrospectively using Pearson correlation between the complete light curves after mapping their observations to common time bins. This similarity is used only for post-hoc analysis and does not enter GP training. Objects are divided into low, medium, and high similarity groups using quantile binning.

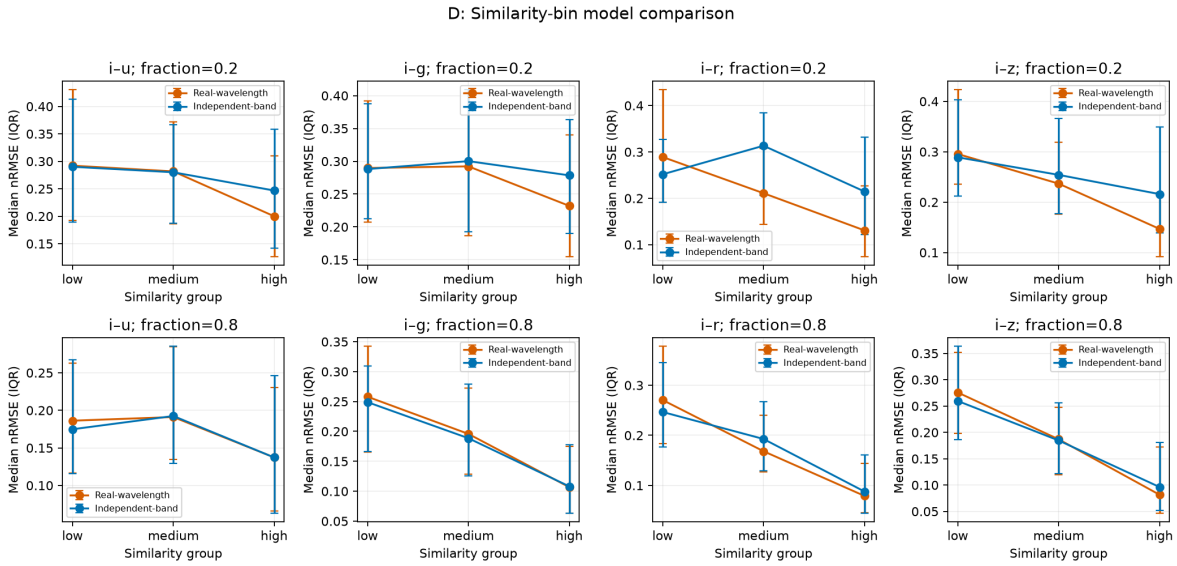
As shown in Figure 12, normalized prediction errors generally decrease with increasing cross-band similarity, particularly for the i–r and i–z pairs. At a target training fraction of 0.2, the real-wavelength GP achieves substantially lower median nCRPS and nRMSE than the independent-band control in the high-similarity groups. In contrast, the differences are relatively small for the more distant u and g bands, even when their empirical temporal similarity is high. When the target training fraction increases to 0.8, the performance differences between the two models become considerably smaller across all auxiliary bands.

Figure 13 further confirms that the benefit of cross-band covariance depends on auxiliary-band choice and target sampling density. At a target fraction of 0.2, the median paired improvements in nCRPS are approximately 0.024 and 0.022 for the i–r and i–z pairs, respectively, substantially larger than those obtained with u or g. The corresponding nRMSE improvements are also greatest for these nearby bands, particularly r. At a target fraction of 0.8, the improvements become much smaller. The fraction of objects benefiting from cross-band covariance in nCRPS is also highest for r and z, reaching approximately 80% and 74%, respectively, under sparse target observations.

These findings are consistent with the wavelength-dependent kernel, which assigns stronger prior covariance to spectrally closer bands. They suggest that temporal similarity is associated with lower reconstruction errors, while the magnitude of cross-band transfer benefits also depends on wavelength separation and target-band data availability. However, differences in observational noise, flux amplitude, and light-curve properties may also contribute to the observed patterns.



(a) Median nCRPS across similarity groups.



(b) Median nRMSE across similarity groups.

Figure 12: Comparison of independent-band and real-wavelength GPs across low, medium, and high empirical cross-band temporal similarity groups for different auxiliary bands and target training fractions. Points represent median performance, and error bars indicate the interquartile range (IQR).

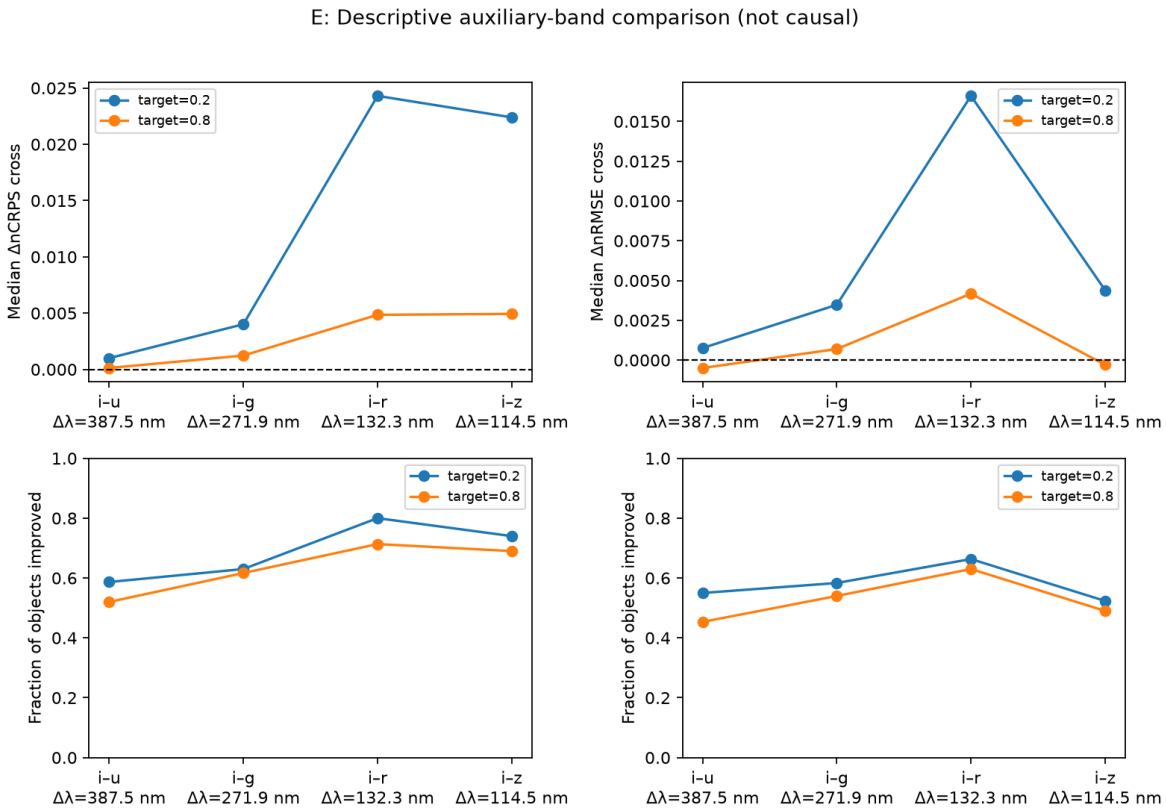


Figure 13: Comparison of auxiliary-band contributions to target i-band reconstruction. The upper panels show median paired improvements in nCRPS and nRMSE, defined as independent-band minus real-wavelength performance. The lower panels show the fractions of objects with positive improvements. Wavelength separations from the target band are indicated below each auxiliary pair.

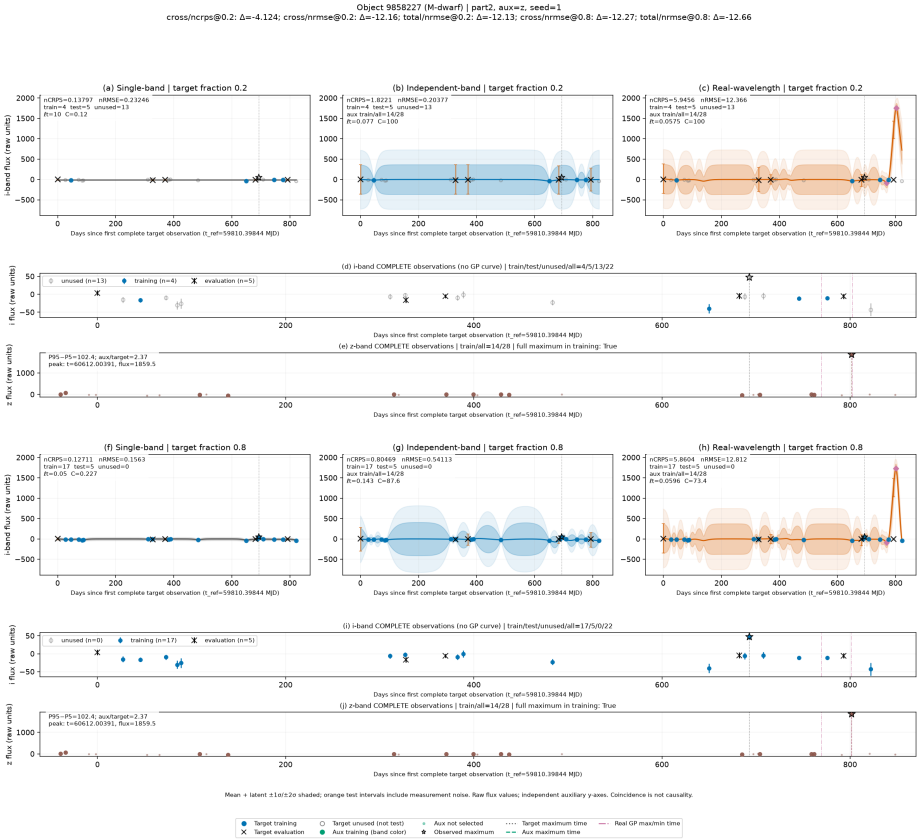
5.2.3 Failure Cases

To better understand the limitations of the multi-band GP, we investigate individual objects for which the real-wavelength GP performs substantially worse than the single-band baseline. For each target training fraction (0.2 and 0.8) and each evaluation metric, we identify the objects exhibiting the largest performance degradation. We focus on configurations with a single auxiliary band to facilitate interpretation of cross-band information transfer.

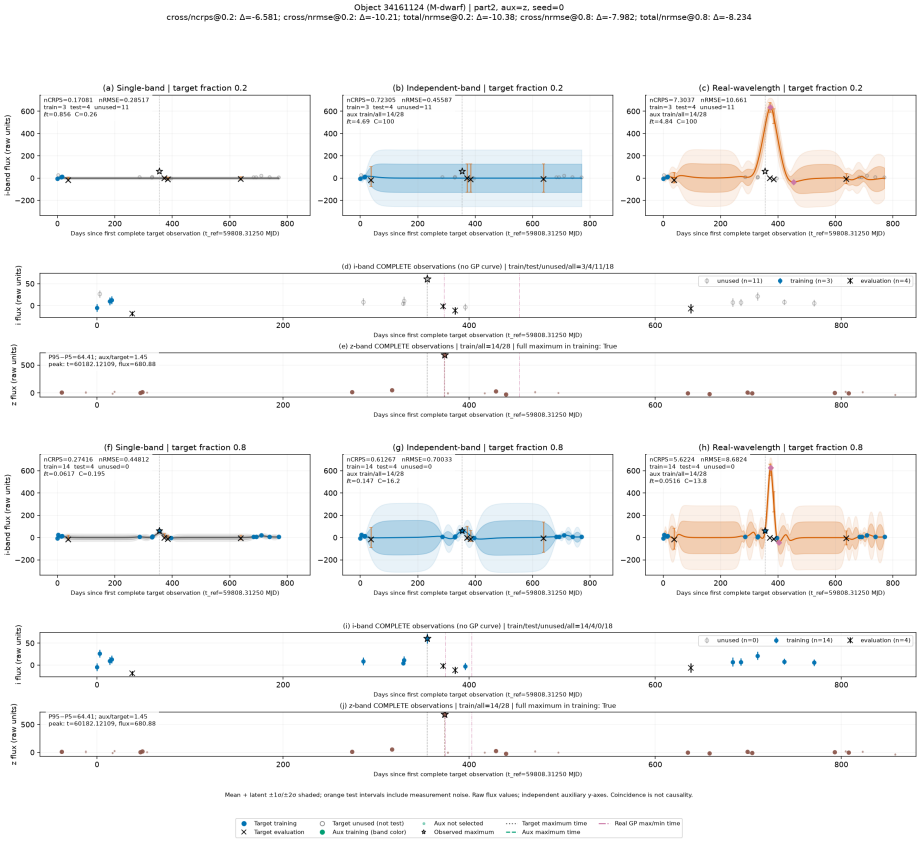
Figure 14 presents two representative failure cases. The posterior means and their $\pm1\sigma$ and $\pm2\sigma$ uncertainty intervals are shown alongside the training and evaluation observations. A recurring pattern among the worst-performing objects is a substantial difference in flux amplitude between the auxiliary and target bands. In Figure 14a, the auxiliary z-band exhibits a strong peak, while the target i-band measurements remain close to zero. Nevertheless, the real-wavelength GP predicts an extremely large peak in the target band, suggesting that the model transfers the auxiliary-band variability without adequately accounting for the difference in amplitude between bands.

A related but distinct failure mode is illustrated in Figure 14b. Here, the real-wavelength GP predicts a peak at approximately the same time as the observed target-band maximum, but substantially overestimates its amplitude. In other cases, the predicted peak is excessively narrow, resulting in large errors when evaluation observations near the predicted peak remain relatively small. These cases suggest that cross-band information may help identify the timing of transient features without reliably recovering their amplitude or shape.

Overall, these examples highlight a limitation of the real-wavelength GP: when the auxiliary and target bands exhibit substantially different flux scales or temporal profiles, cross-band correlations may lead to false predictions. This motivates investigating whether explicitly accounting for band-dependent amplitudes can improve the robustness of multi-band predictions.



(a) Object 9858227: an auxiliary-band peak induces an unrealistically large predicted amplitude in the target band.



(b) Object 34161124: the predicted peak timing is approximately correct, but its amplitude is substantially overestimated.

Figure 14: Representative failure cases of the real-wavelength multi-band GP using the i-band as the target and the z-band as the auxiliary band. Predictions are shown for target training fractions of 0.2 and 0.8, together with the complete observations from both bands. The shaded regions represent the posterior $\pm 1\sigma$ and $\pm 2\sigma$ intervals.

5.3 Band-Specific Amplitude Kernel

The previous failure-case analysis revealed that the real-wavelength GP occasionally produces excessively large target-band predictions associated with large peaks in auxiliary bands. One possible explanation is the shared covariance amplitude in the original separable time–wavelength kernel:

$$k\big((t,b),(t',b')\big) = C\, k_t(t,t')\, k_\lambda(\lambda_b,\lambda_{b'}), \tag{22}$$

where C is a shared covariance amplitude. Since both kernel components have unit diagonal, all bands have the same marginal prior variance, regardless of their observed flux variability.

To relax this assumption, we introduce a band-specific amplitude kernel:

$$\boxed{k\big((t,b),(t',b')\big) = \sigma_b\sigma_{b'}\, k_t(t,t')\, k_\lambda(\lambda_b,\lambda_{b'})} \tag{23}$$

where $\sigma_b > 0$ is the covariance standard-deviation amplitude for band b. Consequently,

$$\mathrm{Var}[f_b(t)] = \sigma_b^2, \qquad \mathrm{Cov}[f_b(t), f_{b'}(t')] = \sigma_b\sigma_{b'} k_t(t,t') k_\lambda(\lambda_b,\lambda_{b'}). \tag{24}$$

This modification allows different bands to have different marginal prior variances while preserving the original temporal and wavelength correlation structures. Both k_t and k_λ remain Matérn-3/2 kernels, with the wavelength length scale fixed at 6000 Å. Only the shared temporal length scale and band-specific amplitudes are optimized. The amplitude parameters are constrained to $\sigma_b \in [10^{-2}, 10^2]$ in internally normalized flux units.

5.3.1 Single-object diagnostic

We first examine PLAsTiCC object 9858227 (M-dwarf), which exhibited severe negative transfer in the previous analysis. The auxiliary z band contains a prominent peak of approximately 1859.5 flux units, while the observed target i-band fluxes remain comparatively small. The original real-wavelength GP transfers this strong auxiliary-band feature into the target prediction, producing a large peak that is not supported by the available target-band observations.

To investigate whether band-specific amplitudes can mitigate this behavior, we compare four models: single-band GP, independent-band GP, original real-wavelength GP with shared amplitude, and the band-specific amplitude GP. We retain the original sampling split, using a target training fraction of 0.2, an auxiliary z-band training fraction of 0.5, and random seed 1. All models use the same preprocessing and evaluation protocol.

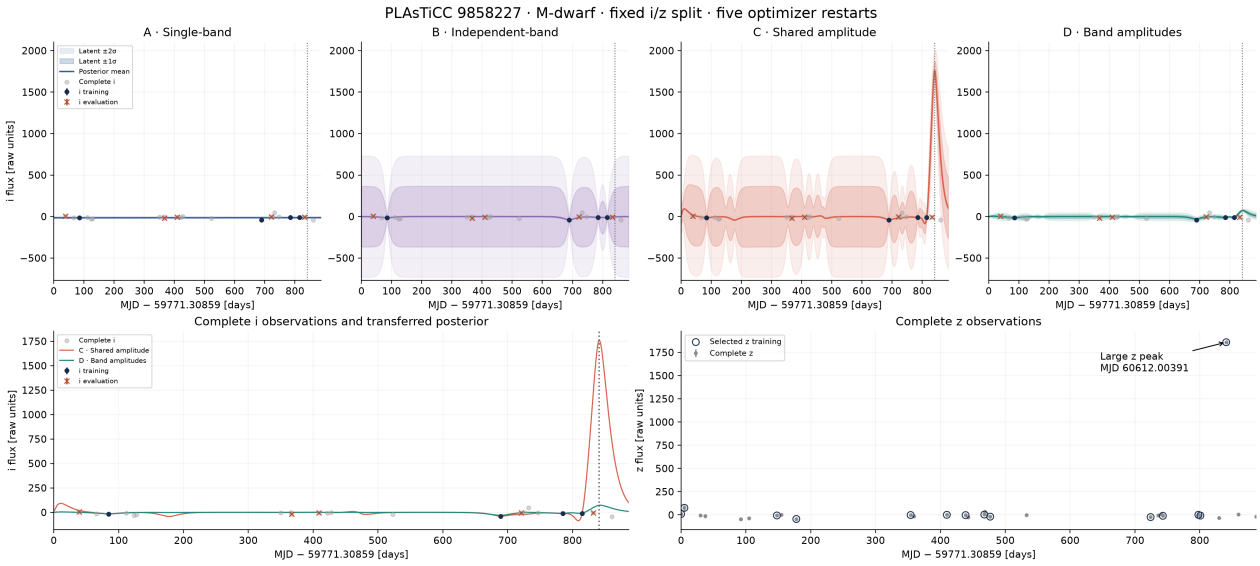


Figure 15: Exact diagnostic for object 9858227 with the original $i + z$ split at target fraction 0.2. Left: full posterior range. Right: zoomed target-flux region with clipped uncertainty. The matched-bounds shared-amplitude model still transfers a large auxiliary-band peak into the target band, while the band-specific amplitude model strongly reduces this behavior.

The learned band-specific amplitudes are $\sigma_i = 0.634$ and $\sigma_z = 15.125$, giving $\sigma_i/\sigma_z = 0.0419$. Neither amplitude reaches its optimization bounds. This substantial amplitude difference is consistent with the

observed discrepancy in flux variability between the two bands. Importantly, the modified kernel preserves the wavelength-dependent correlation coefficient but rescales the covariance magnitude, allowing the target band to respond with a smaller amplitude to auxiliary observations.

Table 5: Reconstruction performance for object 9858227 with target training fraction 0.2. Positive improvements are defined relative to the original shared-amplitude real-wavelength GP.

Model	CRPS	nCRPS	nRMSE	NLPD
Single-band	6.108	0.138	0.232	3.805
Independent-band	80.663	1.822	0.204	6.758
Shared amplitude	263.212	5.946	12.366	9.453
Band amplitudes	13.666	0.309	0.575	5.340

As shown in Table 5, the band-specific amplitude GP reduces nCRPS from 5.946 to 0.309 and nRMSE from 12.366 to 0.575 relative to the original real-wavelength GP, corresponding to reductions of approximately 94.8% and 95.3%, respectively. This represents a substantial improvement in both probabilistic and point prediction errors.

Nevertheless, the modified model still performs worse than the single-band GP, which achieves nCRPS of 0.138 and nRMSE of 0.232. It also improves nCRPS but not nRMSE relative to the independent-band control. Therefore, although band-specific amplitudes substantially reduce the extreme transferred peak, they do not fully resolve the reconstruction difficulties for this object.

A representative success case is shown in Figure 16. For this object, both multi-band models recover the target-band structure better than the single-band and independent-band baselines, but the original shared-amplitude model with matched bounds already performs well. The band-specific amplitude model yields a similar reconstruction, without providing a clear additional gain. This illustrates the other side of the trade-off: when cross-band transfer is already informative, increasing kernel flexibility may weaken the beneficial sharing effect.

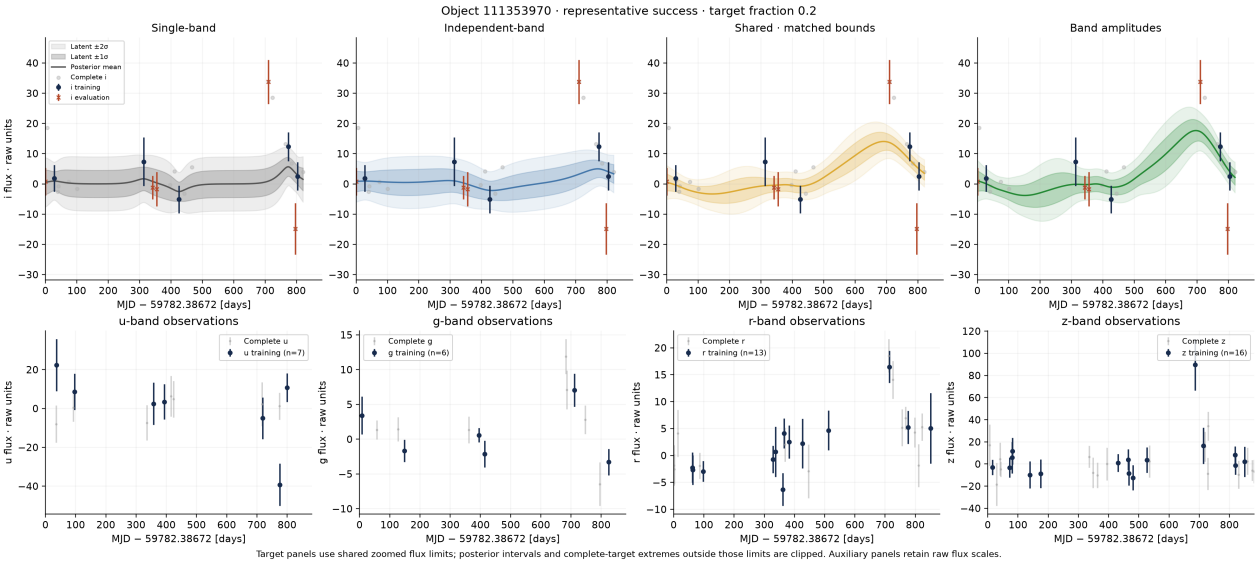


Figure 16: Representative success case using all four auxiliary bands (object 111353970) at target training fraction 0.2. Both multi-band models outperform the single-band and independent-band baselines, but the band-specific amplitude model does not show a clear advantage over the matched-bounds shared-amplitude model.

Overall, this diagnostic demonstrates that relaxing the equal-marginal-variance assumption can substantially mitigate an extreme cross-band amplitude-transfer failure and some normal cases. However, the result concerns only two selected object with very few target evaluation observations and cannot establish whether the modified kernel improves reconstruction performance across the broader population. A population-level comparison is therefore required to assess its generalization and its effect on objects where the original real-wavelength GP already performs well.

5.3.2 300-object experiment

To assess whether the band-specific amplitude kernel improves performance beyond this individual failure case, we evaluate four matched GP configurations on 300 objects: single-band GP, independent-band GP, shared-amplitude multi-band GP with matched variance bounds, and band-specific amplitude multi-band GP. The comparison is performed at target training fractions of 0.2 and 0.8, averaging over three random seeds per object. Performance is summarized using object-level mean nCRPS and nRMSE with equal object weighting.

Figure 17 shows the aggregate results. At target fraction 0.2, both multi-band models substantially outperform the single-band and independent-band baselines in terms of median nCRPS and nRMSE, confirming again that wavelength-dependent information sharing is particularly beneficial when the target band is sparse. However, the shared-amplitude model with matched bounds achieves the best overall performance, with lower median nCRPS and lower median nRMSE than the band-specific amplitude model. The same pattern remains at target fraction 0.8, although the performance differences between all four models become smaller as more target-band observations are available.

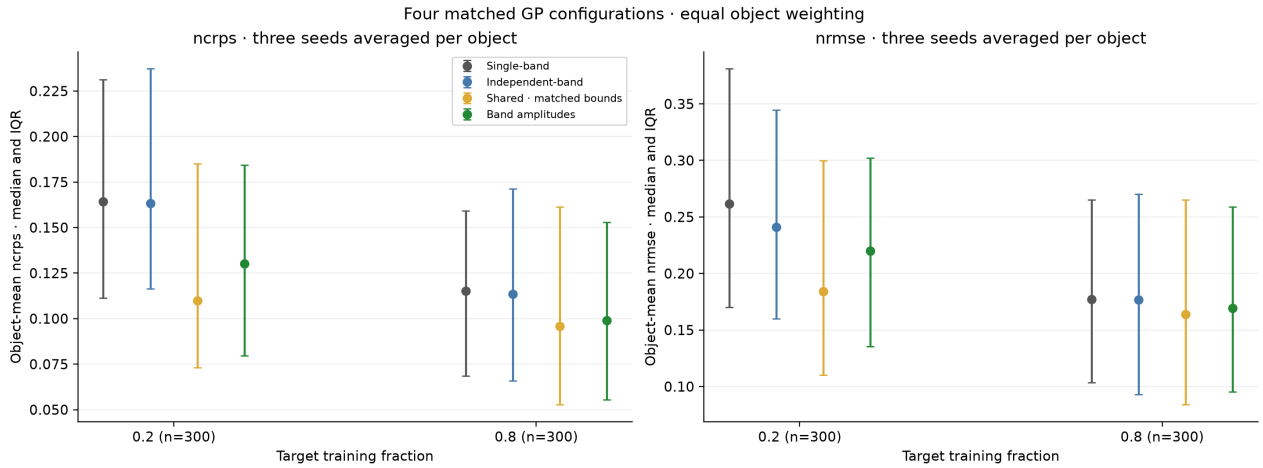


Figure 17: Aggregate performance of four GP configurations across 300 objects. Left: object-level mean nCRPS median and IQR. Right: object-level mean nRMSE median and IQR. Results are shown for target training fractions 0.2 and 0.8.

These results indicate that the additional flexibility of band-specific amplitudes does not translate into a general improvement at the population level. Instead, it appears to trade off robustness to extreme failure cases against the ability to exploit beneficial cross-band transfer in objects where the original multi-band model already works well.

To understand this trade-off more clearly, we divide objects into two groups according to the original shared-amplitude real-wavelength GP:

$$G_{\text{success}} : \text{nCRPS}_{\text{real}} < \text{nCRPS}_{\text{single}},$$
$$G_{\text{failure}} : \text{nCRPS}_{\text{real}} \geq \text{nCRPS}_{\text{single}}.$$

The band-specific amplitude model improves nCRPS for approximately 86% of objects in G_{failure}, but for only about 31% of objects in G_{success}. Thus, the modified kernel is effective primarily as a repair mechanism for failure cases.

This implies that the band-specific amplitude kernel reduces several extreme errors and mitigates some of the most severe negative-transfer outliers. Nevertheless, the worst-case errors remain non-negligible, and the overall distribution of errors is not sufficiently improved to surpass the original MOGP model in median performance.

Overall, the 300-object experiment shows that band-specific amplitudes can mitigate some harmful cross-band transfers, especially in objects that previously suffered from severe negative transfer. However, the modification does not improve the overall median predictive performance, and in many previously successful cases it partially sacrifices useful cross-band information sharing. This suggests that the main challenge is not only to increase kernel flexibility, but to balance flexibility against statistical reliability under sparse observations.

The limited overall benefit may reflect the difficulty of estimating additional covariance parameters from sparse target-band data. Moreover, band-specific amplitudes only rescale covariance magnitudes; they do not address differences in temporal morphology, peak timing, or wavelength-dependent evolution. Future

work could therefore investigate regularized band-specific amplitudes, hierarchical shrinkage toward a shared amplitude, more flexible multi-output covariance structures, or adaptive model-selection strategies that decide when stronger or weaker cross-band sharing is appropriate.

In summary, the band-specific amplitude kernel can mitigate negative transfer caused by differences in flux amplitudes between the target and auxiliary bands. However, the present results suggest that greater kernel flexibility does not necessarily lead to better predictive performance. The key challenge is to preserve the benefits of cross-band information transfer while preventing unrealistic peak propagation and ensuring stable covariance parameter estimation.

5.4 Limitations and Future Work

This study primarily investigates the predictive accuracy and uncertainty calibration of single-band and multi-band GPs, with an emphasis on identifying the factors associated with poor calibration and negative cross-band transfer. Although several possible explanations have been identified, a number of limitations remain.

First, the current analysis mainly examines associations between predictive performance and light-curve properties, including sampling gaps, temporal variability, and cross-band similarity. These relationships do not necessarily establish the underlying causes of miscalibration. More controlled experiments are needed to isolate the effects of observation sparsity, noise, and differences in temporal morphology between bands.

Second, the band-specific amplitude kernel introduces additional covariance parameters, which may be difficult to estimate reliably when target-band observations are sparse. Although this modification substantially reduces errors in many negative-transfer cases, it does not improve the overall median performance compared with the original shared-amplitude kernel. Future work could investigate regularization to allow band-specific flexibility while retaining the benefits of shared covariance structure. An adaptive approach that adjusts the degree of cross-band information sharing according to the available observations and their similarity may also be promising.

Finally, band-specific amplitudes only account for differences in covariance magnitude and cannot fully represent differences in peak timing, temporal morphology, or wavelength-dependent evolution. More flexible multi-output covariance structures could be investigated to capture these relationships. However, their additional complexity should be evaluated carefully, particularly in terms of parameter stability and predictive uncertainty calibration.

6 Conclusion

This study investigated the effectiveness and limitations of wavelength-dependent Gaussian Processes for multi-band light-curve reconstruction, with particular emphasis on predictive uncertainty calibration, target-band data sparsity, cross-band similarity, and covariance kernel design.

The single-band GP experiments revealed substantial differences in predictive performance across astronomical object classes. In particular, supernova-like objects generally exhibited underconfident predictive distributions, while objects with more irregular light curves showed less consistent calibration. The analysis also highlighted the importance of training observations around sharp peaks, where insufficient temporal support can lead to large prediction errors and poorly calibrated uncertainty estimates.

The multi-band experiments demonstrated that wavelength-dependent cross-band covariance generally improves reconstruction accuracy and probabilistic performance, particularly when target-band observations are sparse. The real-wavelength GP achieved lower median nCRPS and nRMSE than the single-band and independent-band baselines across the examined target training fractions. Further analysis showed that spectrally close and temporally similar auxiliary bands were generally associated with greater reconstruction benefits, whereas the advantages of cross-band information diminished as more target-band observations became available. Nevertheless, none of the models achieved perfect uncertainty calibration.

Despite these overall improvements, substantial negative transfer was observed for some individual objects. Failure-case analysis revealed that large differences in flux amplitude between auxiliary and target bands could lead to unrealistic peak predictions. To address this issue, a band-specific amplitude kernel was introduced to relax the equal-variance assumption of the original covariance structure. Although this modification substantially reduced several extreme prediction errors and improved performance for many previously unsuccessful objects, it did not outperform the original shared-amplitude kernel in overall median performance.

Overall, the results demonstrate that wavelength-dependent information sharing is valuable for reconstructing sparse astronomical light curves, but its effectiveness depends on the relationships between bands and the available target observations. Increasing kernel flexibility can mitigate certain negative-transfer failures, but may also weaken beneficial cross-band information sharing. Therefore, a key direction for future

research is to develop more adaptive and statistically reliable covariance models that balance cross-band information transfer, model flexibility, and uncertainty calibration.

7 AI Usage Declaration

In the preparation of this work, ChatGPT was utilized to assist with code development and to refine the language and drafting of the text. All generated outputs were carefully reviewed and validated by the author, who take full responsibility for the final content of this report.

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